Omar AboElAzm

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About

Computer Engineering undergraduate (Senior Year) with a strong interest in embedded systems, hardware/software co-design, and cybersecurity. Hands-on experience in VHDL-based hardware design, Python/C programming, and real-world engineering projects. Actively developing skills in infrastructure security, vulnerability assessment, and penetration testing, with a growing focus on securing embedded and networked systems. Driven by curiosity and a strong learning mindset, I continuously explore, research, and experiment with emerging technologies to become an efficient and security-aware hardware engineer.

Education

Bachelor of Engineering in Computer Engineering
El Shorouk Academy, Cairo, Egypt
2021 – Present (graduation: 2026)

Technical Skills

- Programming Languages: C, Java, C#, Python, Futter, Dart
- Hardware Design: VHDL, Verilog (basic understanding)
- Tools & Frameworks: Git, OpenCV, TensorFlow, Linux (fundamental usage)
- Web Development: PHP, WordPress, MySQL, HTML/CSS
- Design: Adobe Photoshop, Adobe Illustrator, Fusion360

Projects

- **Automatic Indoor Lighting Control System using VHDL (HDL Designer)**
  Designed and implemented a light control system that maintains a constant room brightness and automatically opens and closes curtains. Developed on an FPGA board using VHDL, the system integrates sensor input with logic control to improve energy efficiency and user comfort.
- **Face and Object Detection System using Luckfox Pico Mini SBC**
  This lightweight vision system detects faces and objects in real time. The design was tailored for low-power, resource-constrained hardware using the Luckfox Pico Mini SBC.
- **Wearable Bracelet Communication System based on ESP32 and BLE**
  With a design focused on accessibility and user interaction, this ESP32-based bracelet is touch-sensitive and communicates with a mobile app via BLE. It provides haptic and visual feedback and enables real-time alerts between connected individuals.

Future Projects & Research Interests

Currently planning and working on and conceptualizing several advanced embedded systems and AI-based solutions with a focus on low-power and small hardware, real-time interaction, small size, and practical problem-solving. Areas of interest are:

- A visually impaired wearable assistant equipped with AI for real-time detection of objects and obstacles, with forehead haptic feedback for guiding user, and hybrid processing across embedded device and mobile device, supported with offline LLM.
- ESP32 with LVGL interface small device with modular application system for support of install/uninstall dynamic apps via mobile Bluetooth, app state persistence, and SD card.
- Bluetooth mesh platform for smart environments that has support for decentralized device-to-device communication and dynamic scene management.

Experience

- System Administrator (Part-Time, Remote) | EC4SDF – Cairo, Egypt | 2023 – Present
  Managing and maintaining organizational servers and websites, handling Linux hosting environments, WordPress infrastructure and basic security hardening.

- Freelance Software Developer (Remote) | 2022 – Present
  Utilizing Python and Java for software development, providing optimal solutions, and collaborating with remote teams.

Courses & Certifications

- Vulnerability Analyst & Penetration Tester – DEPI, *In Progress – Expected July 2026*
- ML, CV, NLP (Upscaling Program) – NTI, 2025
- HCIA-Datacom – Huawei Talent Academy, 2025
- HCIA-AI – Huawei Talent Academy, 2024
- IoT Application Development – ITI, 2024
- Mobile Application Development: Flutter, Dart – Udemy, 2023

Activities

- TEDxSHA – Graphic Design team member.
- Enactus – Presentation team member.
- Participating in Game Development Hackathons organized by ITI.

Languages

- Arabic: Native
- English: Fluent
- German: Intermediate